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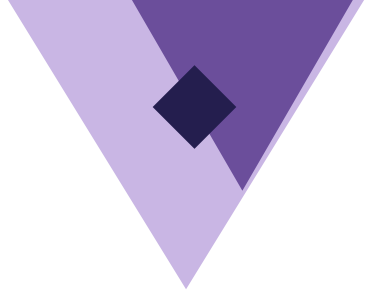
Basic Algebraic Properties of Complex Numbers



Class XII — Mathematics • Chapter 2: Complex Numbers



Introduction



The properties of addition and multiplication of complex numbers behave exactly the same way as they do for real numbers.

In other words — whatever algebraic rules you already trust for real numbers (like $3 + 5 = 5 + 3$) continue to hold true when the numbers are complex.

Under Addition

- Closure
- Commutative
- Associative
- Additive Identity
- Additive Inverse

Under Multiplication

- Closure
- Commutative
- Associative
- Multiplicative Identity
- Multiplicative Inverse

(i) Closure Property

Addition

For any two complex numbers z_1 and z_2 ,
the sum $z_1 + z_2$ is also a complex number.

Multiplication

For any two complex numbers z_1 and z_2 ,
the product $z_1 z_2$ is also a complex number.

In short: adding or multiplying two complex numbers can never “escape” the set of complex numbers – the result always stays complex.

(ii) Commutative Property

Addition

For any two complex numbers

z_1 and z_2 ,

$$z_1 + z_2 = z_2 + z_1$$

Multiplication

For any two complex numbers

z_1 and z_2 ,

$$z_1 z_2 = z_2 z_1$$

In other words: the order in which you add (or multiply) two complex numbers does not matter — swapping them gives the same result.

(iii) Associative Property

Addition

For any three complex numbers

z_1, z_2 and z_3 ,

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

Multiplication

For any three complex numbers

z_1, z_2 and z_3 ,

$$(z_1 z_2) z_3 = z_1 (z_2 z_3)$$

In other words: when adding (or multiplying) three complex numbers, it does not matter which pair you group together first – the result stays the same.

(iv) The Identity Element

Addition

There exists a complex number $0 = 0 + 0i$
such that for every complex number z ,

$$z + 0 = 0 + z = z$$

The complex number $0 = 0 + 0i$ is known as the additive identity.

Multiplication

There exists a complex number $1 = 1 + 0i$
such that for every complex number z ,

$$z \cdot 1 = 1 \cdot z = z$$

The complex number $1 = 1 + 0i$ is known as the multiplicative identity.

(v) The Inverse Element

Addition

For every complex number z , there exists a complex number $-z$

such that,

$$z + (-z) = (-z) + z = 0$$

$-z$ is called the additive inverse of z .

Multiplication

For any nonzero complex number z , there exists a complex number w

such that,

$$zw = wz = 1$$

w is called the multiplicative inverse of z , and is denoted by z^{-1} .

(vi) Distributive Property

Multiplication distributes over addition.

For any three complex numbers z_1 , z_2 and z_3

$$z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3$$

$$(z_1 + z_2) z_3 = z_1 z_3 + z_2 z_3$$

This is the same distributive law that connects addition and multiplication of real numbers — multiplying a sum is the same as multiplying each term separately and then adding.

Proof — Commutative Property of Addition

Claim: $z_1 + z_2 = z_2 + z_1$

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, where $x_1, x_2, y_1, y_2 \in \mathbb{R}$.

$$z_1 + z_2 = (x_1 + iy_1) + (x_2 + iy_2)$$

$$= (x_1 + x_2) + i(y_1 + y_2)$$

$$= (x_2 + x_1) + i(y_2 + y_1) \quad \text{since } x_1, x_2, y_1, y_2 \text{ are real numbers}$$

$$= z_2 + z_1$$

This proves the commutative property under addition. Similar proofs verify the remaining properties.

Proof — Existence of the Multiplicative Inverse (Part 1)

The multiplicative inverse is less obvious than the additive one, so we derive it directly.

Let $z^{-1} = u + iv$ be the inverse of $z = x + iy$.

We must have $z z^{-1} = 1$.

$$\text{That is, } (x + iy)(u + iv) = 1$$

$$\Rightarrow (xu - yv) + i(xv + uy) = 1 + i0$$

Equating real and imaginary parts:

$$xu - yv = 1 \quad \text{and} \quad xv + uy = 0$$

Proof — Existence of the Multiplicative Inverse (Part 2)

Solving the two simultaneous equations in u and v :

$$\mathbf{u = x/(x^2+y^2) \quad v = -y/(x^2+y^2)}$$

(since z is non-zero $\Rightarrow x^2 + y^2 > 0$, so division is always valid)

So, if $z = x + iy$, then:

$$\mathbf{z^{-1} = x/(x^2+y^2) + i \cdot (-y)/(x^2+y^2)}$$

- z^{-1} is not defined when $z = 0$.
- In practice we write $z^{-1} = 1/z$ for convenience.
- For $z_2 \neq 0$, the quotient of z_1 and z_2 is denoted z_1/z_2 .

Complex Numbers Obey the Laws of Indices

For any nonzero complex number z and integers m, n :

$$(i) \quad z^m \cdot z^n = z^{m+n}$$

$$(ii) \quad z^m / z^n = z^{m-n} \quad (z \neq 0)$$

$$(iii) \quad (z^m)^n = z^{mn}$$

$$(iv) \quad (z_1 z_2)^m = z_1^m z_2^m$$



End of Section 2.3

Basic Algebraic Properties of Complex Numbers

